
Streaming Sequence-to-Sequence Learning with Delayed Streams Modeling

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Abstract

We introduce Delayed Streams Modeling (DSM), a flexible formulation for streaming, multimodal sequence-to-sequence learning. Sequence-to-sequence generation is typically cast in an offline manner: the model consumes the complete input sequence before generating the first output timestep. DSM instead models time-aligned streams with a decoder-only language model. By further introducing delays between streams, and selectively feeding or sampling them, DSM provides streaming inference of arbitrary output sequences, from any input combination, making it applicable to many sequence-to-sequence problems. In particular, given a text and audio stream, automatic speech recognition (ASR) corresponds to the text stream being delayed, while the opposite gives a text-to-speech (TTS) model. We perform extensive experiments for these two major sequence-to-sequence tasks, showing that DSM provides state-of-the-art performance and latency while supporting arbitrary long sequences, being even competitive with offline baselines. We demonstrate DSM applications on <https://delayed-stream-modeling.github.io/>

1 Introduction

We are interested in streaming sequence-to-sequence (seq2seq) learning, i.e. predicting an output sequence as we process an input sequence synchronously, as opposed to offline seq2seq where inputs are recorded entirely before producing the output sequence. The latter class of offline models was introduced for a diverse set of tasks such as handwriting recognition (Graves et al., 2013), automatic speech recognition (ASR) (Graves et al., 2013) or machine translation (Bahdanau et al., 2015; Sutskever et al., 2014), by designing modality-dependent input encoders, typically coupled with a text decoder (Hochreiter & Schmidhuber, 1997). Although this asymmetry between input processing and output generation facilitated the adoption of this framework in many tasks, it also led to a divergence of model architectures across modalities. As an example, a Tacotron text-to-speech (TTS) model (Wang et al., 2017) would differ from an ASR model such as LAS (Chan et al., 2016). The advent of decoder-only Transformers (Vaswani et al., 2017) for text language modeling reduced the gap between input and output processing by allowing a single model to process a simple concatenation of tokens. In parallel, neural compression algorithms that can transform images (Razavi et al., 2019; Esser et al., 2020) and audio (Zeghidour et al., 2022; Défossez et al., 2023) into discrete tokens analogous to text allowed integrating these modalities along text sequences. Thus, a decoder-only model can be used for seq2seq tasks such as ASR (Rubenstein et al., 2023), TTS (Wang et al., 2023), spoken dialogue (Défossez et al., 2024), visual understanding (Beyer et al., 2024) or image generation (Ramesh et al., 2021). Furthermore, inputs and outputs are interchangeable in this framework, meaning a single model can be trained for generation in both directions: AudioPALM (Rubenstein et al., 2023) performs TTS and ASR, while CM3Leon (Yu et al., 2023) provides both image captioning and generation. Yet, a major limitation of these decoder-only

approaches is their incompatibility with streaming. First, their prefix-based formulation requires access to the full input sequence before generation, which prevents real-time inference and inherently limits the maximum input length. Second, modalities operate at differing framerates: audio or video tokens are typically sampled regularly, while text tokens represent linguistic units pronounced over varying durations. This prevents applications such as meeting transcription or continuous translation.

In this work, we present Delayed Streams Modeling (DSM), a framework for streaming sequence-to-sequence learning across modalities. DSM uses a decoder-only model to process as many parallel token streams as there are I/O sequences. This multistream architecture, introduced by Défossez et al. (2024), allows for a synchronous autoregressive modeling of aligned sequences which—when coupled with a finite context—provides real-time, streaming generation over infinite input sequences. Moreover, by operating at a constant framerate, DSM allows for batching, a feature rarely provided by streaming models. The second key component of DSM is the introduction of a delay between streams to control the quality/latency trade-off: shifting a sequence B such that it is delayed w.r.t. sequence A allows for a better prediction of the former based on the latter. With an appropriate masking, a DSM model can be trained to continuously predict any combination of output sequences from any combination of input sequences. To illustrate the abilities of the DSM framework, we train speech-text models for ASR and TTS. We show how DSM provides a state-of-the-art tradeoff between latency—as low as a few hundred milliseconds—and quality, while providing long-form synthesis and transcription, along with precise word timestamps that locate where they are pronounced. We furthermore introduce *delay conditioning* which allows controlling the quality/latency trade-off at inference time, without retraining a model. We will release our code and models, along with an evaluation dataset for long-form dialog TTS.

2 Related Work

Streaming Sequence-to-Sequence Learning. Most streaming seq2seq literature has focused on speech-to-text tasks, in particular ASR (Li et al., 2021) and translation (Zhang et al., 2024; Barrault et al., 2023). Monotonic (Raffel et al., 2017; Chiu & Raffel, 2018) and local (Chiu et al., 2019) attention respectively allow for causal attention of outputs with respect to inputs along and handling arbitrarily long sequences. A common limitation of streaming models is their incompatibility with batching when using an inference policy (Barrault et al., 2023), or the lack of symmetry meaning that specific models must be used for speech-to-text (Li et al., 2021) and text-to-speech (Wang et al., 2017). In contrast, DSM allows for batching and accelerated inference. In the context of this paper, this allows DSM to be trained for state-of-the-art ASR or TTS (see Figure 1), as shown in Section 4, with its performance being even competitive with offline approaches.

Multimodal language models. Transformer-based autoregressive models are the current main approach to sequence-to-sequence problems. They were introduced by Vaswani et al. (2017) for machine translation, and were soon extended to multimodal tasks, such as ASR (Radford et al., 2023) or visual understanding (Alayrac et al., 2022), by designing modality-specific encoders. More recently, neural codecs have provided compact, discrete representations of images (Esser et al., 2020) and audio (Zeghidour et al., 2022) that remove the need for modality-specific encoders inside the generative model, while providing a symmetrical processing of inputs and outputs which allows performing bidirectional tasks (e.g. speech-to-text and text-to-speech (Rubenstein et al., 2023)) with a single architecture. Défossez et al. (2024) introduce a multistream decoder architecture for spoken dialogue, which predicts text and audio tokens in a streaming fashion, later applied by Labiausse et al. (2025) to real-time speech translation. In this work we extend the approach of Défossez et al. (2024), in order to reach state-of-the-art performance on the two most competitive speech-text tasks, namely ASR and TTS. Moreover, while (Défossez et al., 2024) and (Labiausse et al., 2025) operate with a delay specified before training, we propose delay conditioning for inference-time latency control without retraining.

3 Method

Notation. We wish to solve a sequence-to-sequence task between two domains $\mathcal{X}$ and $\mathcal{Y}$. Each domain consists of sequences of vectors of all possible lengths, e.g.

$$\mathcal{X} = \bigcup_{T \in \mathbb{N}} \{(X_t) \in \mathbb{R}^{T \times d}\}, \quad \mathcal{Y} = \bigcup_{T' \in \mathbb{N}} \{(Y_{t'}) \in \mathbb{R}^{T' \times d'}\}. \quad (1)$$

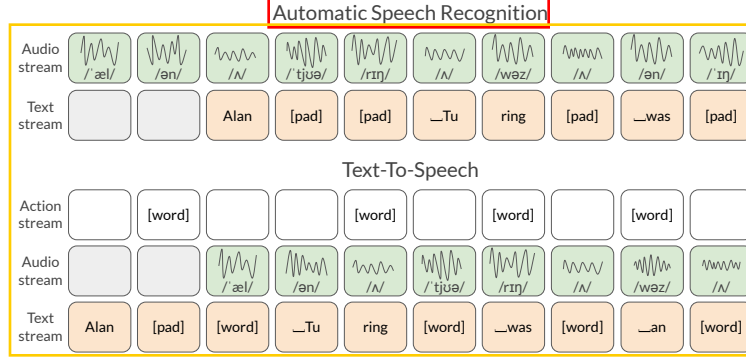


Figure 1: Delayed stream modeling for speech-text tasks. Depending on which stream is delayed with respect to the other, we solve either an ASR or a TTS task. For TTS, we further need an action stream for the model to let us know when it is ready to receive a new word.

In the case where either X_t or Y_t is discrete-valued, we can use a one-hot representation for it in Eq. (4). We assume that we are given a joint probability distribution over the outer product domain $\mathcal{X} \times \mathcal{Y}$, and that we have the random variables $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$, along with the joint distribution

$$\mathbb{P}[X, Y] = p(X, Y). \quad (2)$$

We also introduce $T \in \mathbb{N}$ (resp. T') the random variable indicating the length of X (resp. Y), along with the marginals $p(X)$ and $p(Y)$. For any sequence Z_t and index t , we denote $Z_{<t} = (Z_1, \dots, Z_{t-1})$, potentially empty if $t < 0$. We similarly define $Z_{<t}$, $Z_{>t}$ and $Z_{\geq t}$.

Sequence-to-sequence as joint modeling. Let's assume for this paragraph that $\mathcal{X}$ is the set of all possible monophonic waveforms sampled at 24 kHz, and $\mathcal{Y}$ is made of sequences of one-hot encoded vectors over a set of words. Intuitively, we assume there exists a coupling $p(X, Y)$ such that $p(X, Y)$ is high if Y represents the transcription of X , or conversely, if X represents a speech utterance of the text given by Y . Formally, the task of ASR corresponds to sampling from the distribution $\mathbb{P}[Y|X]$, while the task of TTS corresponds to sampling from the distribution $\mathbb{P}[X|Y]$. Thus, each task can be solved by accurately estimating both probability distributions,

$$q(X, Y) \approx \mathbb{P}[Y|X], \quad q'(Y, X) \approx \mathbb{P}[X|Y]. \quad (3)$$

For simplicity, we now only focus on estimating $\mathbb{P}[Y|X]$, the inverse task being obtained by exchanging the definition of X and Y . We thus call $\mathcal{X}$ the input domain, and $\mathcal{Y}$ the output domain.

Auto-regressive modeling of Y . A good candidate for estimating $\mathbb{P}[Y|X]$ is auto-regressive modeling, with a Transformer model (Vaswani et al., 2017), under the extra assumptions that the output domain $\mathcal{Y}$ can be discretized. Thus, one would estimate

$$q(y|X, Y_{<t}) \approx \mathbb{P}[Y_t = y|X, Y_{<t}]. \quad (4)$$

One can then sample Y auto-regressively, knowing X . Due to the lack of explicit structure between the time grid t of X and t' of Y , one would usually condition on the entirety of X , e.g. when using Transformer based models, either by prefixing the entire sequence X before the generation Y , or by providing X through cross-attention layers, which is mathematically equivalent. This forbids the use of the model in a streaming fashion, as the entire input signal X must be known ahead of time, and cannot be extended once the generation of Y has started. Such methods often require explicit and manual chunking and stitching operations, which also reduces their ability to be efficiently batched. Conversely, aligning X and Y to the same frame rate allows for batched streaming inference.

Aligning sequences for streaming prediction. We assume that both domains $\mathcal{X}$ and $\mathcal{Y}$ can share the same time grid, e.g. $(X_t) \in \mathbb{R}^{T \times d}$ and $(Y_t) \in \mathbb{R}^{T \times d'}$. We call two such aligned sequences *streams*. Then one can simply model

$$q_{\text{aligned}}(y|X_{<t}, Y_{<t}) \approx \mathbb{P}[Y_t = y|X_{<t}, Y_{<t}]. \quad (5)$$

Given $X \sim p(X)$, we sample auto-regressively from Eq. (5), with a streaming context X ,

$$\tilde{Y}_1 \sim q_{\text{aligned}}(\tilde{Y}_1|X_1), \quad \tilde{Y}_t \sim q_{\text{aligned}}(\tilde{Y}_t|X_{\leq t}, \tilde{Y}_{<t}). \quad (6)$$

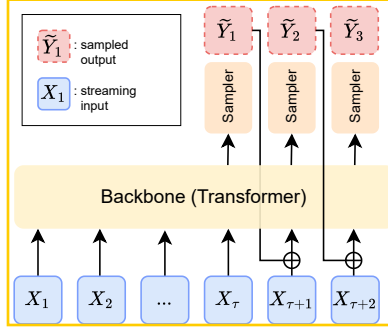


Figure 2: **DSM Architecture.** Transformer is fed with the streaming input X_t . After a delay τ , a sampler is fed with the output of the backbone samples Y_t . At the next step, the backbone receives both the sampled value and next streaming input, whose embeddings are summed.

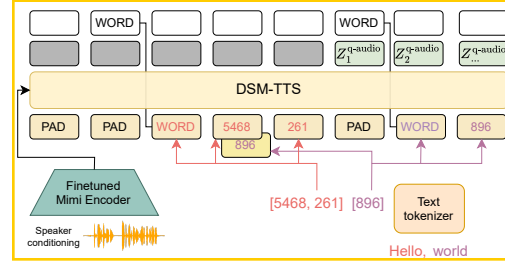


Figure 3: **DSM-TTS inference.** The input words “Hello, world” are tokenized. Until the model action stream outputs a WORD, it is fed with PAD. Then the first word’s tokens are fed, including a look-ahead text stream. Once a delay $\tau = 5$ has accumulated, the model also outputs the audio.

118 We would want that given $X \sim p(X)$, then $(X, \tilde{Y}) \sim (X, Y)$, so that in particular $\mathbb{P}[\tilde{Y}|X] \approx$
 119 $\mathbb{P}[Y|X]$. However this needs not be the case unless certain conditions are met.

120 **The importance of causality.** In particular, for $(X, \tilde{Y}) \sim (X, Y)$ to be true, $Y_{<t}$ must be independent
 121 of $X_{>t}$, knowing $X_{<t}$. To realize that, one can look at a simple counter-example taking $X_t \sim$
 122 $\mathcal{B}(0.5)$ independent Bernoulli variables, and $Y_t = X_t \oplus X_{t+1}$ the XOR of X_t and X_{t+1} . Clearly
 123 $\mathbb{P}[Y_t|X_{<t}, Y_{<t}] \sim \mathcal{B}(0.5)$ for all t , yet, given $X = (0, 1)$, one would have

$$Y_1 = 1 \quad \text{a.s.}, \quad \tilde{Y}_1 \sim \mathcal{B}(0.5).$$

124 Thus $Y|X$ and $\tilde{Y}|X$ have different distributions. Intuitively, given that we do not sample X but
 125 teacher-force real-world data, we must ensure that when sampling $\tilde{Y}_t$, no future value of $X_{>t}$ might
 126 end up in “contradiction” with the value we sampled.

127 **Delaying the output stream.** In practice, this is achieved by delaying the output stream Y_t by a
 128 number of steps $\tau > 0$. Thus, we replace Eq. (5) by

$$q_\tau(y|X_{\leq t+\tau}, Y_{<t}) \approx \mathbb{P}[Y_t = y|X_{\leq t+\tau}, Y_{<t}], \quad (7)$$

129 and define $\tilde{Y}^\tau$, similarly to the procedure described in Eq. (6). Perfect independence is hard to achieve;
 130 in the case of ASR, a named entity might be ambiguous without context, and only future development
 131 in a discussion would resolve this ambiguity. Taking $\tau = T$ recovers the prefixing or cross-attention
 132 approaches presented earlier. In practice, there is a trade-off between the level of independence of Y_t
 133 with $X_{>t+\tau}$, and the latency of the method.

134 **Architecture of a DSM.** In practice, a DSM, depicted in Figure 2, involves three components: (i) an
 135 auto-regressive backbone, (ii) an input embedder for X and Y into the backbone, and (iii) a sampler
 136 for Y^τ conditioned on the output of the backbone. The backbone can be a Transformer architecture,
 137 optionally equipped with cross-attention layers to provide further non-streaming contextual
 138 information. The embedder for X and Y can be learnt embedding tables in the case where both
 139 domains are discrete, which are summed before going into the backbone. On the output side, we mask
 140 the loss on the tokens of X and only compute cross-entropy on Y . Finally, the conditional sampler
 141 can be a linear layer applied to the output of the backbone to derive logits if Y is discrete. It could
 142 also be a flow or diffusion model conditioned on the output of the backbone for the continuous case.

143 3.1 Representations of the speech and text domains

144 We demonstrate the DSM framework on ASR and TTS, where the two domains are text and audio.

145 **Audio.** Given a waveform $m \in \mathbb{R}^{d_s \cdot f_s}$ with the duration in seconds d_s and the sample rate
 146 $f_s = 24 \text{ kHz}$, we turn it into a more compact latent space using the Mimi codec (Défossez et al.,

147 (2024), giving us a sequence of tensors $Z^{\text{q-audio}} \in \mathbb{R}^{d_s \cdot f_r \times d_{\text{audio}}}$ with a frame rate of $f_r = 12.5$ Hz. This
 148 latent space is discretized with Residual Vector Quantization (Zeghidour et al., 2022) (RVQ), giving us
 149 a set of $Q \in \llbracket 1, 32 \rrbracket$ coarse-to-fine discrete values per time step with cardinality $N_a = 2048$, each com-
 150 ing from one codebook in the RVQ, giving a quantized representation $Z^{\text{q-audio}} \in \{1, \dots, N_a\}^{d_s \cdot f_r \times Q}$.
 151 **Text.** We tokenize text using a vocabulary of N_t , specifically trained on speech data transcriptions.
 152 Two tokens have a special meaning: PAD (indicating the absence of words at this time) and WORD (in-
 153 dicating the start of a new word) following Défossez et al. (2024). Given a transcript, with word-level
 154 timestamps, of a waveform of duration d_s , its aligned text representation is $Z^{\text{text}} \in \{1, \dots, N_t\}^{d_s \cdot f_r}$.
 155 For each word in the transcript represented by tokens $(x_1, \dots, x_n) \in \{1, \dots, N_t\}^n$ and starting at
 156 $s \in \mathbb{R}^+$ seconds, we define its start index $i = \text{floor}(s \cdot f_r)$, and store it as $Z^{\text{text}}_i \leftarrow \text{WORD}$, $Z^{\text{text}}_{i+1} \leftarrow x_1$,
 157 $Z^{\text{text}}_{i+2} \leftarrow x_2$, etc. Any step in Z^{text} not assigned by a word token is given the special value PAD.

158 3.2 DSM for automatic speech recognition: DSM-ASR

159 For ASR, we consider $X = Z^{\text{q-audio}}$ and $Y = Z^{\text{text}}$. By predicting the word tokens of Y , we learn to
 160 transcribe audio, while computing the loss on PAD and WORD tokens trains the model to predict the pre-
 161 cise boundaries of each word. At inference time, we teacher-force the audio tokens of X and sample
 162 the full sequence Z^{text} to obtain a transcription along with timestamps with a precision of 80ms (frame
 163 size). This is allowed by the fact that we apply a constant delay to all words in the sequence, meaning
 164 we only need to shift the output timestamps back by the same value to recover the true timestamps.

165 **Deriving aligned speech-text data.** We are looking for fine-grained alignment between speech and
 166 text, however speech datasets are typically aligned at the level of the sentence (Panayotov et al., 2015).
 167 Conveniently, whisper-timestamped (Louradour, 2023) provides automatic transcriptions with
 168 word-level timestamps. We rely on these pseudo-labels for the pretraining phase of DSM-ASR. We
 169 then finetune on a mixture of public datasets with ground-truth transcripts (see details in Section 4.2),
 170 which pose two challenges. First, the automatic transcriptions extracted by Whisper in pretraining are
 171 formatted with capitalization and punctuation, but the level of formatting varies a lot between datasets.
 172 To address this, we train a 300M prefix-LM for automatic formatting, on a dataset of formatted
 173 Whisper transcripts. A second challenge is that these ground-truth transcripts do not have word-level
 174 alignment. We derive those by producing pseudo-transcripts with Whisper, and reconciling them
 175 with the formatted transcript using a Dynamic Time Warping algorithm (Giorgino, 2009).

176 **Delay conditioning for inference-time control.** As shown in Section 4.3.1, transcription quality is
 177 heavily dependent on the delay between audio and text. Thus, training DSM-ASR with a fixed delay
 178 requires choosing a latency/quality trade-off beforehand, and retraining a new model for each delay,
 179 despite the training task remaining fundamentally the same. To instead control this trade-off at infer-
 180 ence, we train DSM-ASR over random delays, sampled for each sequence. The model is additionally
 181 conditioned on a cosine embedding (Vaswani et al., 2017) of the delay (expressed in milliseconds),
 182 added to the inputs. Experiments in Section 4.3.1 show that this delay conditioning performs at least as
 183 well as models with a fixed delay, and that the effective delay precisely respects the conditioning value.

184 3.3 DSM for text-to-speech

185 We further apply DSM to TTS, taking $X = Z^{\text{text}}$, $Y = Z^{\text{q-audio}}$. We use a stream delay of 2s (or 25
 186 steps) on the output audio. For sampling along the Q dimension in $Z^{\text{q-audio}}$, we use a RQ-Transformer
 187 as a sampler (Lee et al., 2022; Défossez et al., 2024), i.e. a smaller Transformer conditioned on
 188 the output of the backbone at each timestep and performing autoregressive modeling along the Q
 189 dimension. All the backbone inputs (generated audio tokens and next word token input) are fed
 190 through learnt embeddings and summed. We are confronted with the problem that the input domain is
 191 no longer plain text, but text properly padded for time alignment. While at train time we can teacher-
 192 force the ground-truth padded text, this is not the case for a novel text to synthesize at inference time.

193 **Action output stream.** We add an extra stream to the TTS outputs, whose goal is to predict whether
 194 the next input text token will be a WORD token or not. This special input token indicates that a new word
 195 is starting, and that its tokens are going to follow as inputs. This extra stream controls an inference-
 196 time *action*: when predicted by the model, we will feed as input the text tokens for the next word over
 197 the next time steps. While these are being fed, the model is not allowed to output another WORD action.
 198 The action output stream is not fed back into the model as it is redundant with the text stream input.

Table 1: **Short-form ASR performance.** We report Word Error Rates (WER, %) for DSM-ASR and selected non-streaming baselines from the OpenASR leaderboard, along with streaming baselines.

MODEL	AVG.	AMI	EARNINGS22	GIGASPEECH	LS CLEAN	LS OTHER	SPGISPEECH	TED-LIUM	VOXPOPULI
<i>Non-streaming</i>									
WHISPER MEDIUM.EN	8.1	16.7	12.6	11.0	3.0	5.9	3.3	4.1	9.6
WHISPER LARGE-V3	7.5	16.0	11.3	10.0	2.0	3.9	2.9	3.9	9.5
CRISPERWHISPER	6.7	8.7	12.9	10.2	1.8	4.0	2.7	3.2	9.8
CANARY-FLASH	6.4	13.1	12.8	9.9	1.5	2.9	2.0	3.1	5.6
PHI-4 MULTIMODAL	6.1	11.5	10.5	9.8	1.7	3.8	3.1	2.9	5.9
PARAKEET-TDT-v2	6.1	11.2	11.2	9.7	1.7	3.2	2.2	3.4	6.0
<i>Streaming</i>									
WHISPER MEDIUM.EN	9.0	22.1	13.4	10.4	3.0	6.2	3.7	4.7	8.6
WHISPER LARGE-V3	9.4	18.4	11.0	10.0	8.4	12.6	3.2	3.8	7.9
DSM-ASR	6.3	11.7	10.6	9.7	1.7	4.3	2.0	3.4	6.8

Lookahead second text stream. The action stream allows the model to predict the next word position, although the model has no knowledge of its content for making that decision. The delay between text and audio only provides context for the audio generation, however, the decision on where to insert pauses and words has no such context. Given a sequence of words $m_1, m_2, \dots$, the lookahead text stream feeds the tokens of the words m_{i+l} to the backbone while the primary text feed contains the tokens of words m_i .

Speaker conditioning. We provide speaker embeddings for up to 5 speakers. Each speaker is represented by a 10s audio extract of the same speaker outside of the training segment. Speakers are identified using the diarization tool Pyannote (Bredin, 2023). One speaker is elected the main speaker. When a turn of the main speaker starts, its first word is prefixed with a special MAIN token, while when any other speaker turn starts, it is prefixed with OTHER. This allows us to generate dialogs with control over change of turns and speaker voices. Each speaker audio extract is encoded with a copy of the Mimi encoder, whose Transformer is fine tuned along with the main TTS model, while convolution layers are kept frozen for stability. We concatenate all the speaker embeddings, sum them with an absolute positional embedding (Vaswani et al., 2017), and feed them through cross-attention layers to the backbone. An overview of the whole DSM-TTS inference process is shown in Figure 3.

4 Experiments

4.1 Architectural hyperparameters

We use a Transformer (Vaswani et al., 2017) backbone with RoPE positional encoding (Su et al., 2024). For the DSM-TTS experiments, we use a 1B parameters backbone with a 2048 dimension latent space, GLU feed-forward units, 16 layers, and 16 heads of attention. The DSM-ASR uses a 3B parameters backbone, with 2048 dimensions, 48 layers, and 32 attention heads, and a linear to predict the logits over the text vocabulary, with a cardinality $N_t^{\text{asr}} = 4000$, trained for English only. The model uses a variable delay τ which is sampled per batch item in a range of [0.25, 4]s (Section 3.2). The TTS model also receives the speaker embedding through cross-attention. The sampler is a Transformer along the codebook Q dimension described in Section 3.1 with no context over the time axis, with a dimension of 1024, 6 layers for each codebook, with a linear layer to estimate the logits. The text tokenizer is trained on bilingual French/English data, with a cardinality $N_t^{\text{tts}} = 8000$. The model uses a delay τ of 2s, or 25 steps, and a lookahead stream with $l = 2$ (Section 3.3). We use AdamW (Loshchilov & Hutter, 2019), a cosine learning rate schedule with linear warmup, with an initial rate of $2 \cdot 10^{-4}$ for the TTS, and $4 \cdot 10^{-4}$ for the ASR, and a weight decay of 0.1.

4.2 Training protocol

Pretraining. We use an audio collection of 2.5 million hours of publicly available audio content in English, and French transcribed with whisper-timestamped. Given the synthetic nature of text transcripts, this phase amounts to hard distillation of whisper-timestamped. We train DSM-ASR on random 90s segments for 1.6M steps, on 48 H100s. DSM-TTS is trained on 120s audio extracts, on 16 H100s, 250k updates with batch size 64.

Table 2: **Long-form ASR performance.** We report Word Error Rates (WER, %) across four long-form datasets for DSM-ASR and a set of streaming and non-streaming baselines.

MODEL	AVG.	TED-LIUM	MEANWHILE	REV16	EARNINGS21
<i>Non-streaming</i>					
DISTIL-LARGE V2	8.7	3.7	7.8	12.2	11.2
WHISPER-LARGE-V2	9.0	4.4	6.3	13.6	11.8
<i>Streaming</i>					
WHISPER MEDIUM-EN	9.0	3.9	6.7	13.0	12.5
WHISPER LARGE-V3	8.1	3.4	6.1	11.4	11.4
DSM-ASR	7.9	2.9	5.7	12.3	10.6

Finetuning (DSM-ASR). We then finetune the model on a collection of public datasets with ground-truth transcripts, described in Appendix A.1 and totaling 24 hours. This training stage lasts for 100k updates with batches of 128 examples, using 16 H100s.

We then adapt the model to long-form inputs, which most public datasets lack, by constructing a long-form mixture described in Appendix A.2. We run this stage for DSM-ASR for 25k updates with batch size 32, using 16 H100s.

4.3 Automatic Speech Recognition

We evaluate DSM-ASR (with a default delay of 2.5s) in terms of transcription quality, latency, and timestamps precision. We consider short-form transcription (shorter than 30s), as it is the focus of the OpenASR Leaderboard (Srivastav et al., 2023). We also look at streaming inference for long-form transcription (up to 2 hours).

Baselines. We benchmark DSM-ASR against leading models of the OpenASR Leaderboard, including Whisper (Radford et al., 2023), Canary-Flash (Zelasko et al., 2025), Phi-4 Multimodal Instruct (Abouelenin et al., 2025) and Parakeet (Xu et al., 2023). Notably, all these models perform non-streaming ASR, as they require access to the full input sequence. We thus also include a streaming variant of Whisper (Macháček et al., 2023), with a delay of 2.5s for a fair comparison. For long-form transcription, we add the Distil-Whisper (Gandhi et al., 2023) variant.

Transcription quality. We report micro-averaged Word Error Rate (WER), which avoids over-emphasizing short sequences, and is the standard computation used in the OpenASR Leaderboard, of which we use the official evaluation codebase.¹ Throughout this Section, we use Whisper normalizer for English (Radford et al., 2023).

Latency. We evaluate the latency of streaming models as the average delay between the real timestamp of a word, and the time when this word is transcribed in the output. In the absence of ground-truth timestamps, we use pseudo-timestamps on Librispeech test-clean (Panayotov et al., 2015) provided by Lugosch et al. (2019). These timestamps were obtained by Montreal Forced Aligner (McAuliffe et al., 2025), and use them as reference.

Timestamps. See Appendix C for a complete description.

4.3.1 ASR Results

Short-form transcription. Table 1 shows that DSM-ASR is significantly better than streaming baselines, and even competitive with the best, non-streaming models of the OpenASR Leaderboard. With an average WER of 6.3%—while 6.1% being the current best score of the leaderboard—DSM-ASR is remarkably the only streaming model among top ASR systems. In Appendix C, we see that, in terms of timestamp precision, DSM-ASR performs significantly better than Whisper Large-v3 while somewhat underperforming CrisperWhisper, though with a better WER.

Long-form transcription. Table 2 reports WER values across 4 long-form datasets with sequences up to 2 hours: TED-LIUM (Hernandez et al., 2018), Meanwhile (Radford et al., 2023), Rev16 (Radford et al., 2023), and Earnings21 (Rio et al., 2021). We see that DSM-ASR outperforms both streaming and non-streaming baselines.

¹https://github.com/huggingface/open_asr_leaderboard

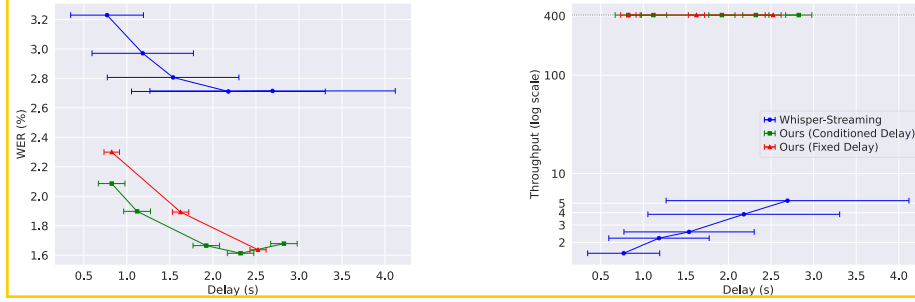


Figure 4: **ASR WER and throughput vs. delay.** Word Error Rate (WER, %) (left) and throughput (right) in function of the delay. Throughput is the product between Real-Time Factor and batch size.

Table 3: **Long-form TTS WER.** We compare open-source and closed-source baselines. Short-form inference is ran with a short context. ElevenLabs is evaluated on a limited subset due to its cost.

MODEL	WER ENGLISH (%) (↓)			WER FRENCH (%) (↓)		
	AVG.	DIALOGS	MONOLOGUES	AVG.	DIALOGS	MONOLOGUES
<i>Short-form inference</i>						
DIA	6.7	7.8	5.7	24.9	22.4	27.3
CSM	4.7	3.9	5.4	-	-	-
ORPHEUS	7.1	3.8	10.5	-	-	-
<i>Long-form inference</i>						
CSM	15.5	15.7	15.4	-	-	-
DSM-TTS (OURS)	3.6	2.8	4.3	6.4	3.7	9.1
<i>Long-form inference, small subset</i>						
DSM-TTS (OURS)	7.5	10.4	4.5	8.1	8.9	7.4
11LABS FLASH v2.5	2.5	1.0	4.0	4.5	4.1	4.9
11LABS MULTILINGUAL V2	2.1	1.0	3.1	2.9	2.8	3.0

Delay conditioning and latency. Figure 4 (left) compares the WER obtained for DSM-ASR with and without delay conditioning, along with Whisper-Streaming. We observe that the delay of Whisper-Streaming has a large variance, while DSM-ASR has a precision of $\sim 300\text{ms}$ around its target delay. Interestingly, training a single DSM-ASR model with delay conditioning outperforms fixed delay variants. Figure 4 (right) shows the throughput on an H100 GPU: DSM-ASR can process 400 sequences simultaneously while being real-time and its throughput is independent of the delay. This is unlike Whisper-Streaming which reduces its delay by re-evaluating the partial input sequence more frequently, increasing the computational cost. Combined with the fact that Whisper-Streaming does not allow for batching, this results in a $100\times$ lower throughput than that of DSM-ASR.

4.4 Text-To-Speech experiments

Evaluation datasets. We collect a novel dataset for long-form TTS evaluation in English and French. We first use news articles from the NTREX-128 (Federmann et al., 2022) text dataset, given 123 monologues per language. To evaluate controllable dialog capabilities, we use 110 synthetic scripts per language generated by a LLM, spanning three categories: daily life, technical, and number-heavy discussions. For voice conditioning, we use samples from the test set of VCTK (Yamagishi et al., 2019) for English, and from the test and valid sets of CML (Oliveira et al., 2023) for French. We provide examples and more details in the Appendix B, and we will release this dataset.

Metrics. We evaluate the per-document WER, using text normalization from Whisper (Radford et al., 2023). We collect subjective metrics covering both the speaker similarity to the conditioning and overall speech quality, see Appendix A for more details.

Baselines. We compare to open-source models Dia, Orpheus, and CSM, as well as the closed-source ElevenLabs API.² Dia and ElevenLabs support French and English, while Orpheus and CSM only support English. Dia, Orpheus and CSM can be speaker-conditioned through prefixing, with Dia

²Available at nari-labs.com/dia, canopy.ai/Orpheus-TTS, [SesameAILabs/csm](https://SesameAILabs.com/csm) and elevenlabs.io

Table 4: **Subjective evaluations on TTS.** We compare with baselines over two axes through human evaluations: speech quality, measured as MUSHRA scores (1–100, along with std. of the mean), and speaker-similarity win-rates, summarized as ELO scores (with 95% confidence intervals).

MODEL	ENGLISH		FRENCH	
	QUALITY (↑)	SPK. SIM. (↑)	QUALITY (↑)	SPK. SIM. (↑)
DIA	52.9 ± 2.0	1930.0 ± 22.6	30.0 ± 2.0	1578.7 ± 53.8
CSM	43.0 ± 1.4	2056.0 ± 18.6	-	-
ORPHEUS	33.7 ± 1.8	1820.5 ± 32.7	-	-
DSM-TTS (OURS)	50.5 ± 1.8	2066.9 ± 22.8	52.1 ± 1.6	2078.5 ± 18.0
11LABS FLASH V2.5	64.1 ± 1.7	1977.8 ± 22.6	65.7 ± 1.8	1926.0 ± 15.7
11LABS MULTILINGUAL V2	61.3 ± 1.8	2067.4 ± 23.5	68.7 ± 1.6	2099.3 ± 17.8

Table 5: **TTS: Latency and throughput.** We compare the inference performance of DSM-TTS and selected baselines. Real-Time Factor (RTF) is higher than 1 if the model can produce audio in real time. Throughput is the product between Real-Time-factor and batch size.

MODEL	MODEL SIZE	LATENCY (MS)(↓)	RTF(↑)	THROUGHPUT(↑)
DIA	1.6B	-	0.7	0.7
CSM	1.5B	-	1.0	1.0
ORPHEUS	3.8B	-	0.7	0.7
DSM-TTS B.S.=1	3.7B	185	2.7	2.7
DSM-TTS B.S.=32	3.7B	560	2.1	67.4
DSM-TTS B.S.=64	3.7B	708	1.7	111.3

and CSM supporting dialogs. For Orpheus and ElevenLabs, dialogs are emulated by concatenating single-speaker turns. Details of how baselines are evaluated are provided in Appendix E.

4.4.1 TTS Results

Main results. As seen in Table 3, our approach provides the lowest WER across all languages for both monologues and dialogs. Our method is the only one to run long-form inference across all cases, CSM showing strong degradation when running with longer sequences, Dia only being trained for 20s output, and ElevenLab requiring per-turn generation for dialogs. In terms of subjective results, our model is on par with the commercial ElevenLab models for speaker similarity, outperforming all existing methods. In quality, it is equivalent to the best open-source baseline for English and much better for French, while lagging behind commercial models. Note that we kept all methods with their original sample-rate (e.g. 44.1kHz for ElevenLab) which contributes to the difference.

Throughput and latency. Our method is easily batchable, leading to gains in throughput while staying compatible with real-time generation. As shown in Table 5, on a single H100 the amount of audio generated is 100x real-time, more details are provided in Appendix G.

DSM-ASR and DSM-TTS as a speech interface for LLMs. We combine DSM-ASR, DSM-TTS, and Gemma 3 (Gemma Team et al., 2025) into an LLM voice chat application with sub-second latency. We provide conversation samples at <https://delayed-stream-modeling.github.io/>

5 Conclusion

We introduce Delayed Streams Modeling, a flexible framework for streaming sequence-to-sequence learning. DSM provides a remarkable trade-off between quality and latency, and an unprecedented throughput among streaming models. Focusing on speech-text tasks, DSM-ASR is the first streaming ASR model to provide timestamped, formatted transcripts that competes with the top offline models, while DSM-TTS is competitive with non-streaming baselines while being the only model providing arbitrarily long synthesis. In future work, we will extend DSM to more sequential multimodal tasks.

Limitations. We acknowledge that streaming naturalistic speech with voice conditioning opens up both opportunities in inclusive human-machine interactions and risks of fraudulent impersonation. Addressing the latter requires that public access to such technologies is accompanied by proper user terms, voice verification mechanisms, and watermarking of generated content. Finally, the need for aligned domains reduces the amount of gold-standard ground-truth data that can be used for training.

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